

Lesson 3 · Practice sheet

NCEA Level 2 · AS 91262 · Sketching gradient functions

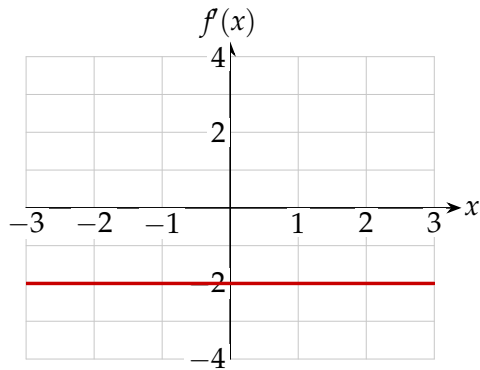
Time: 30 minutes

Answer version · tutor's working, not an official mark scheme

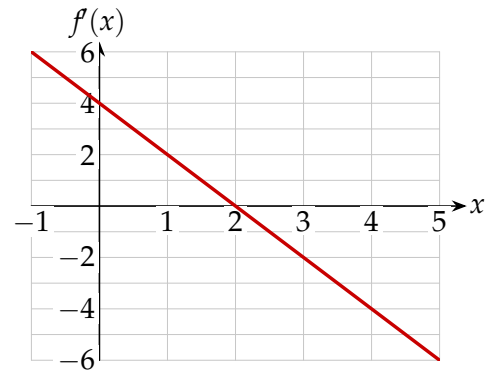
Sketch page (for the practice rounds during the lesson)

Grids 1 to 4: sketch $f'(x)$ for the graph on the screen. Grids 5 and 6: sketch a possible $f(x)$.

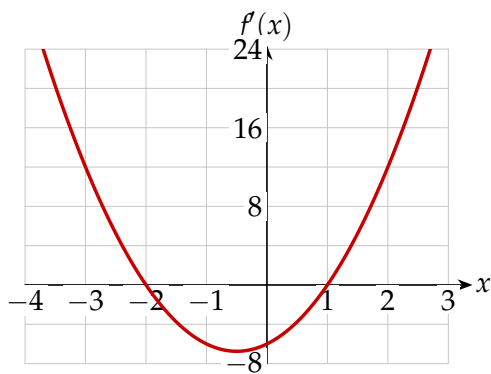
Grid 1



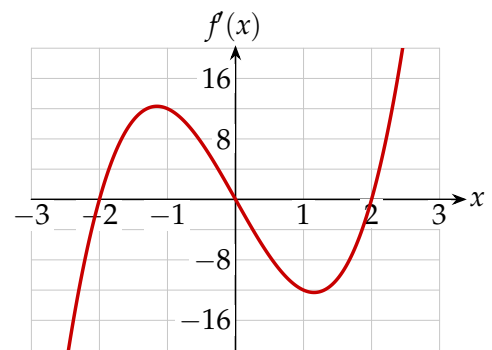
Grid 2



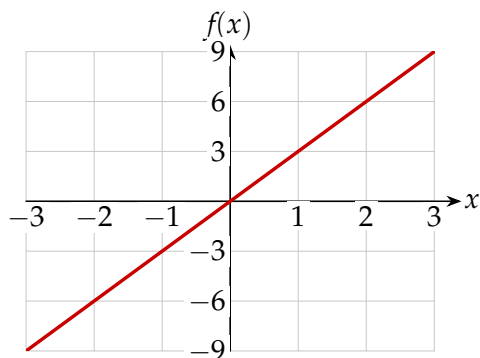
Grid 3



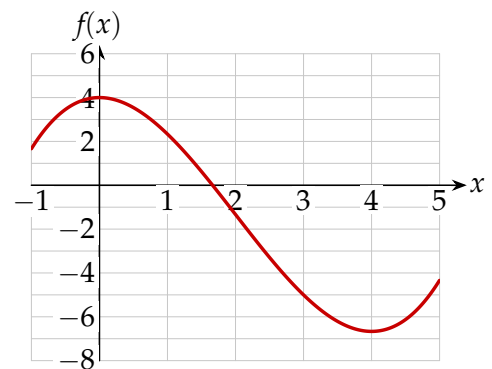
Grid 4



Grid 5



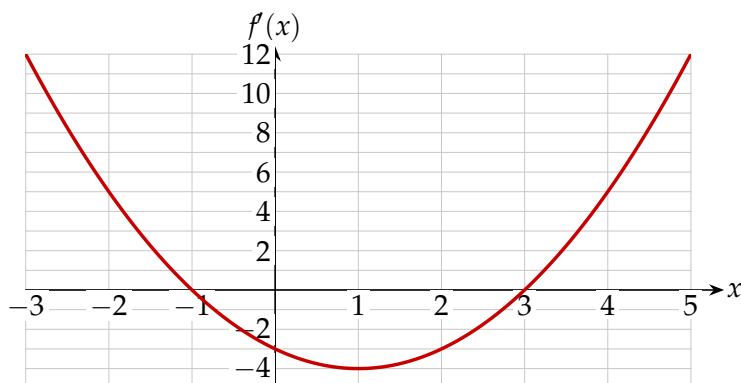
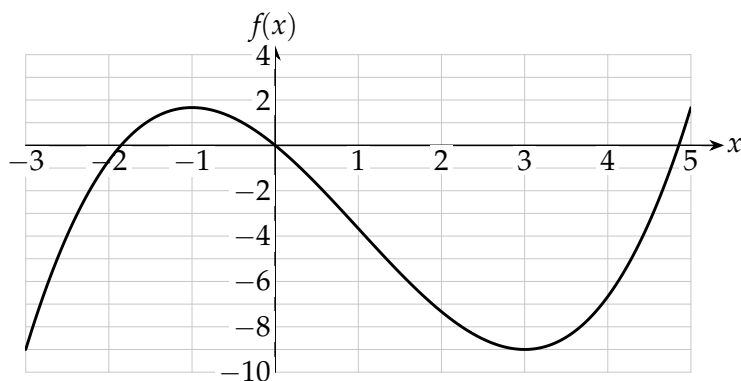
Grid 6



Work alone for 30 minutes. Line up every key x -value with a ruler. [A] Achieved, [M] Merit, [E] Excellence.

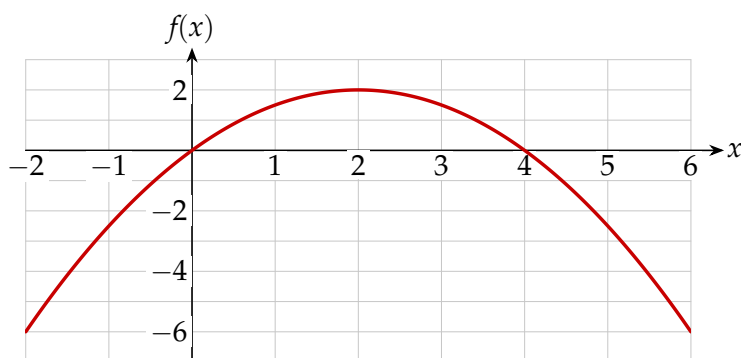
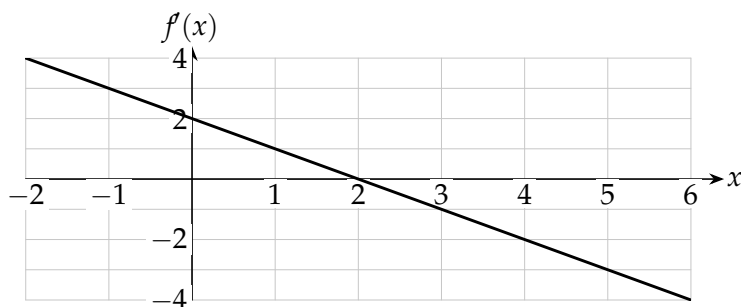
1. The graph of $y = f(x)$ is shown. Sketch the graph of $y = f'(x)$ on the axes below it.

[A]



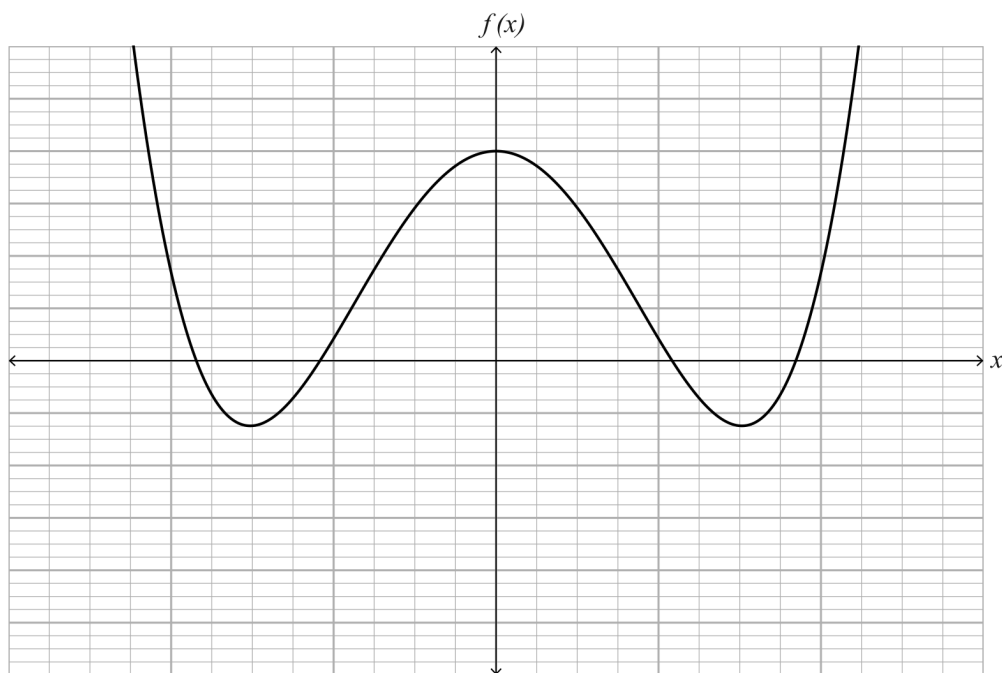
2. The graph of $y = f'(x)$ is shown. Sketch a possible graph of $y = f(x)$ on the axes below it.

[A]

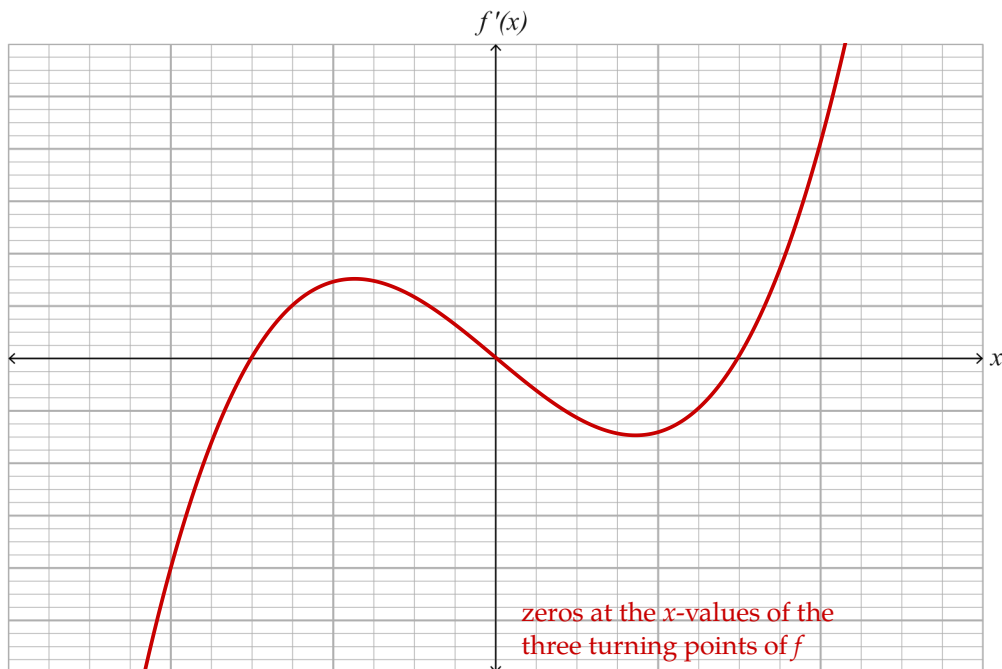


Question 3 [A] · Merit for a fully correct cubic

- (a) The graph of a function $y = f(x)$ is shown on the axes below.

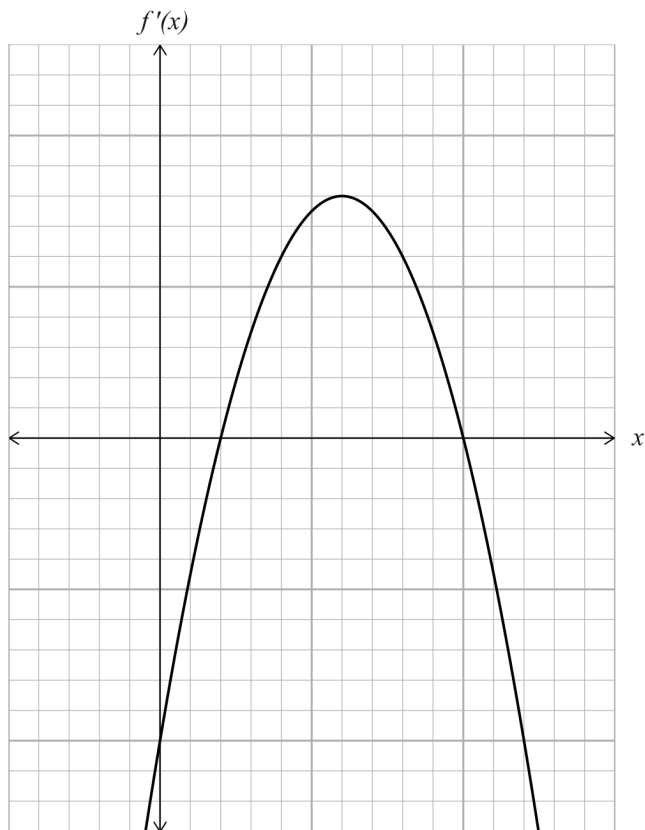


Sketch the graph of the gradient function $y = f'(x)$ on the axes below.
Both sets of axes have the same horizontal scale.

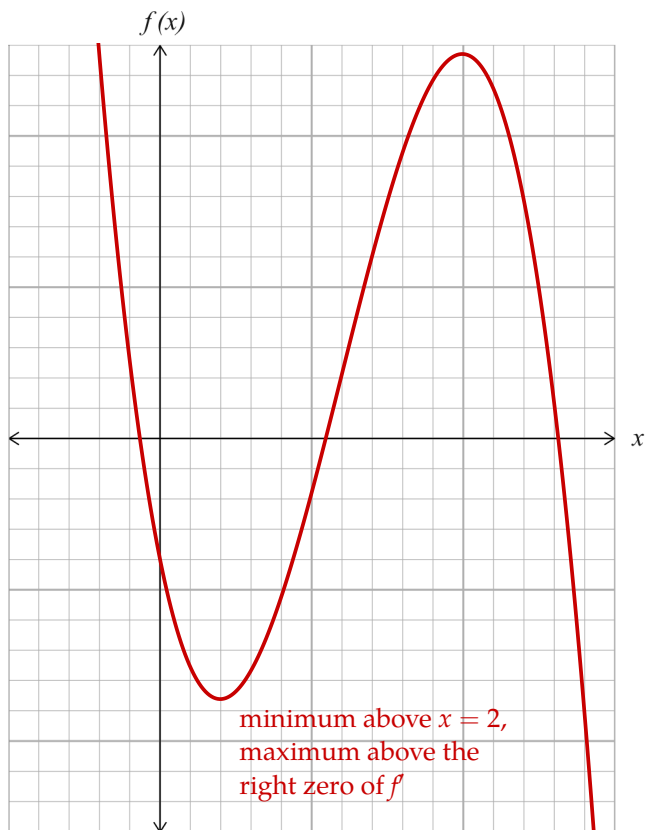


Question 4 [M]

- (c) The diagram below shows the graph of a gradient function $f'(x)$.



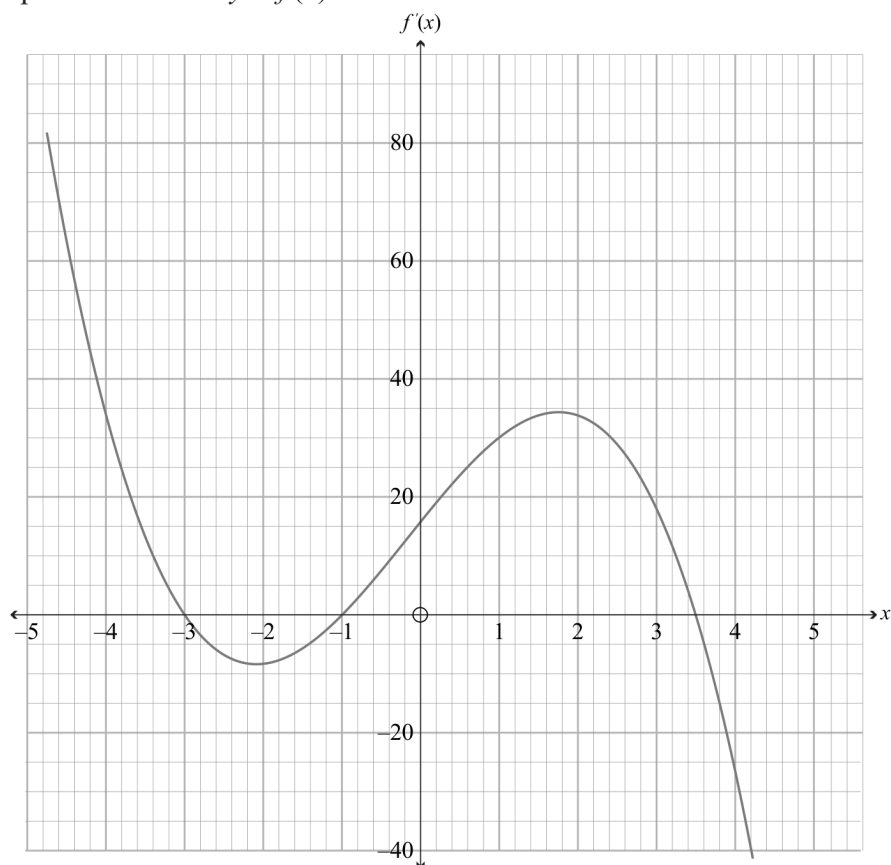
On the axes below sketch the graph of the function $f(x)$.



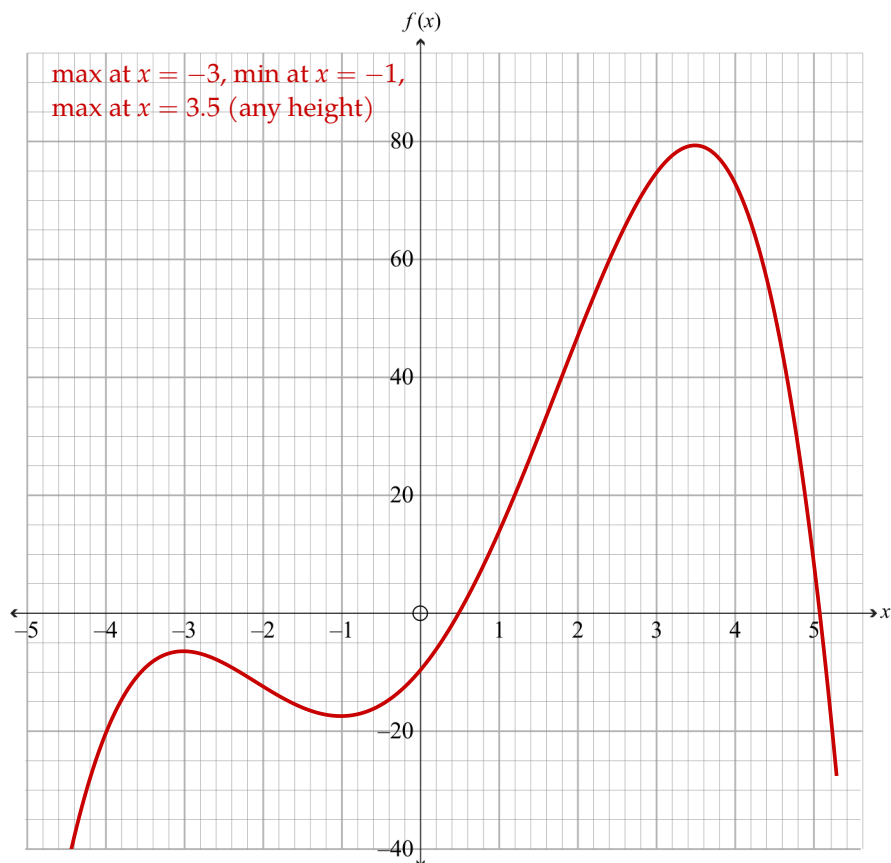
- (a) The gradient graph of a function $y = f(x)$ is shown on the axes below.

Question 5

[M]

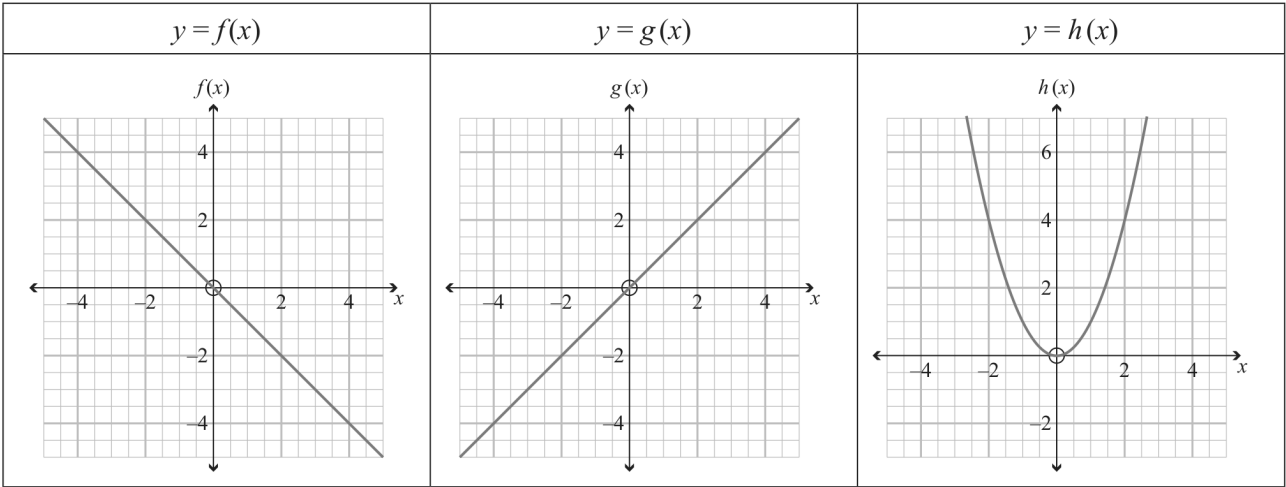


Sketch a possible graph of the original function $y = f(x)$ on the axes below.

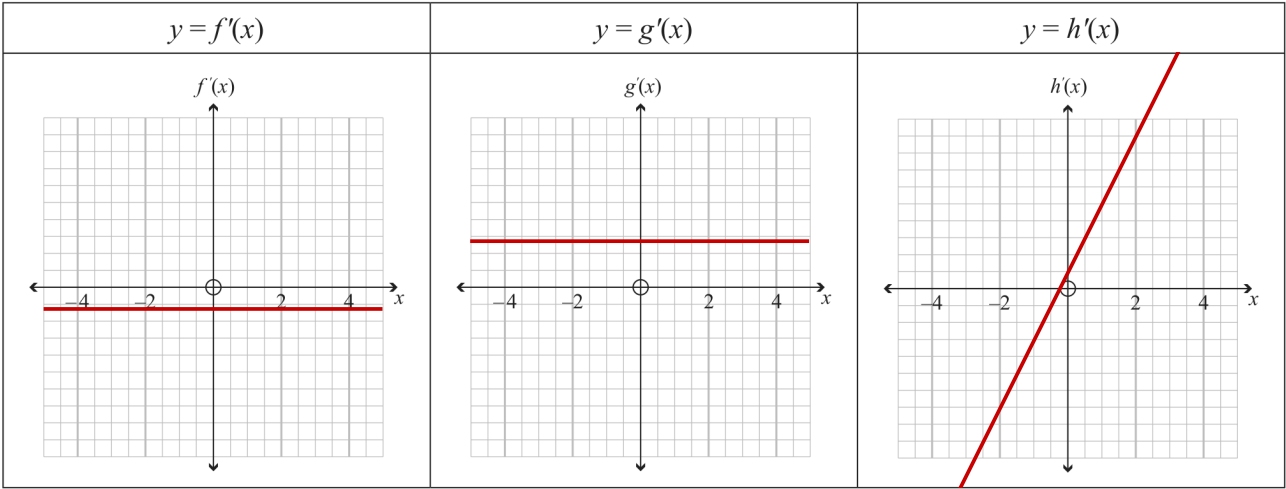


Question 6 (i) [A]

(b) The diagrams below show the graphs for the functions of $y = f(x)$, $y = g(x)$, and $y = h(x)$.

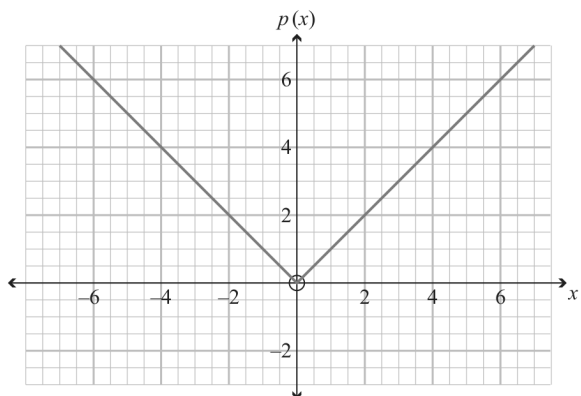


(i) On the axes below, sketch the graphs of their respective gradient functions.

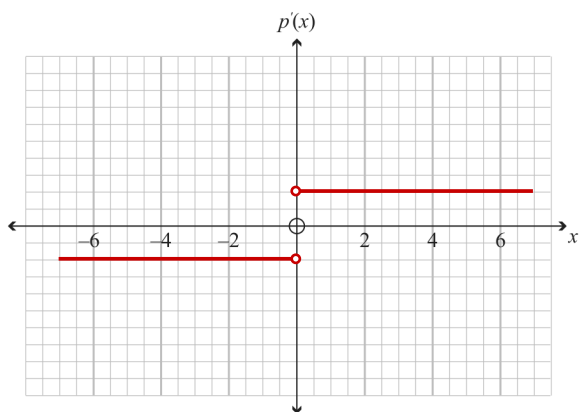


Question 7 (ii) [A] · (iii) [M/E]

(ii) The diagram below shows the graph of the function $y = p(x)$.



On the axes below sketch the graph of the gradient function $y = p'(x)$.



(iii) Is it true to say that the gradient of the function $p(x)$ is zero when $x = 0$?

Justify your response using the graphs on pages 8 and 9, and/or mathematical reasoning.

No. p has a sharp corner at $x = 0$.

Left of 0 the gradient is -1 ; right of 0 it is 1 .

The gradient jumps at $x = 0$, so it is not defined there: it is not 0.

(The open circles on $p'(x)$ show there is no value at $x = 0$.)